ASSET TRACKING SYSTEM

Ultra-Wideband (UWB) Real-Time Indoor Localization Project

PROJECT TEAM

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System Architecture: UWB Tag & Anchor

Two-Way Ranging (TWR) System Overview

1. STM32 Anchor

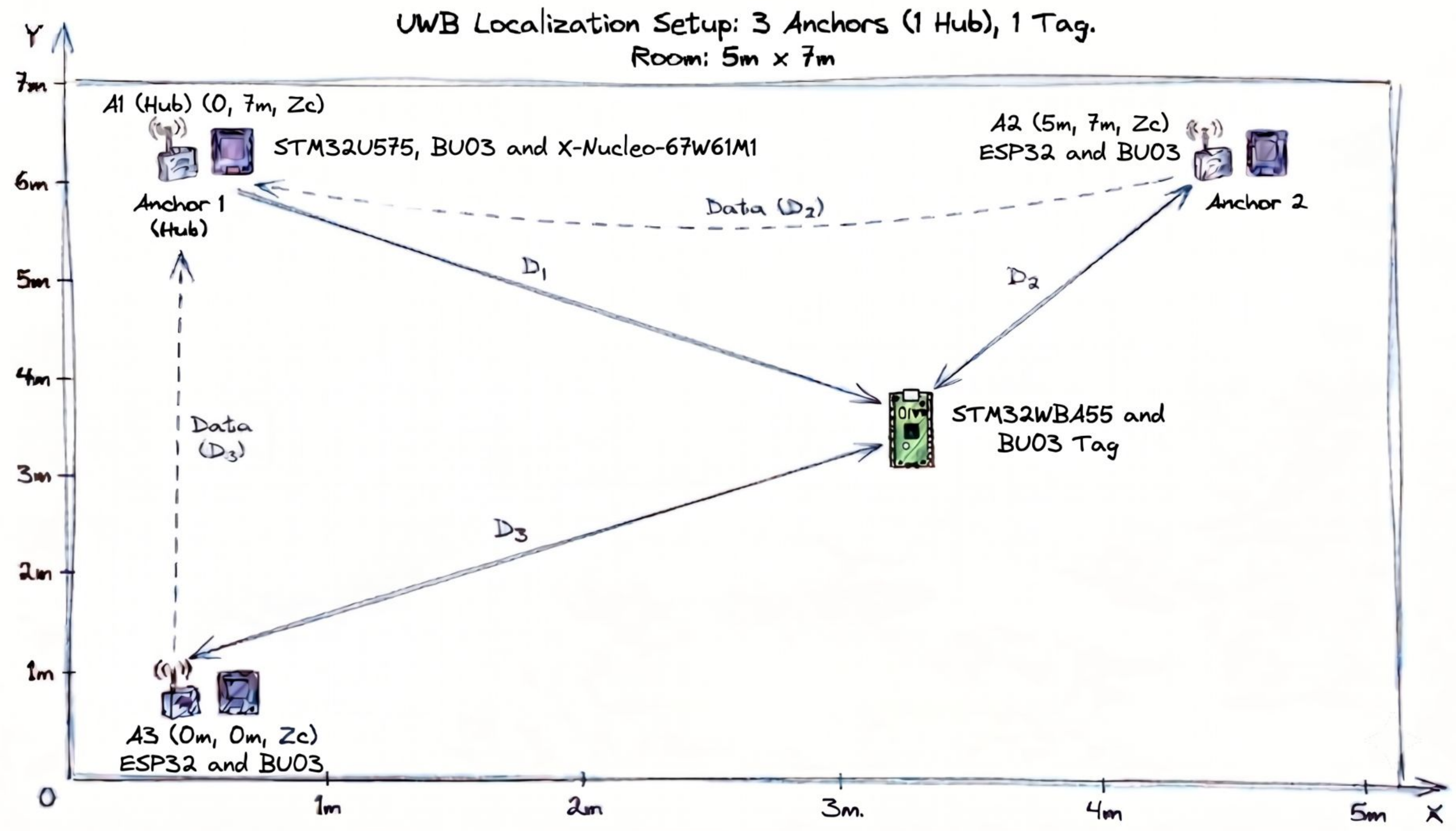
- Anchor node based on the **STM32** platform.
- Equipped with a **BU03 UWB** transceiver module and an **X-NUCLEO Wi-Fi** expansion board.
- Part of the **Asymmetric Two-Way Ranging** (TWR) architecture.

2. STM32 Tag

- Tag built on the **STM32WBA55** MCU.
- Equipped with the companion **BU03 UWB** module.
- Features high-speed **SPI interface** up to **3.125 MBits/s**.

3. Timing & Communication

- Uses **Poll-Response** mechanism on **UWB Channel 5** at **6.8 Mbps**.
- Tag implements a strict, deterministic response delay.
- Transmits exactly **900 microseconds** after the poll broadcast.


UWB Localization Setup: 3 Anchors (1 Hub), 1 Tag.
Room: 5m x 7m
Y
7m
6m
5m
4m
3m
2m
1m
0
1m
2m
3m.
4m
5m
X
A1 (Hub) (0, 7m, Zc)
STM32U575, BU03 and X-Nucleo-67W61M1
Anchor 1
(Hub)
A2 (5m, 7m, Zc)
ESP32 and BU03
Anchor 2
Data (D2)
D1
D2
Data
(D3)
STM32WBA55 and
BU03 Tag
D3
A3 (0m, 0m, Zc)
ESP32 and BU03

Hardware Specifications

Tag, Anchor, and BU03 Transceiver profiles

TAG NODE HARDWARE

System Platform

STMicroelectronics NUCLEO-WBA55CG

MCU: STM32WBA55CGUx

CORE: Cortex-M33 @ 100MHz

OS: Baremetal (No RTOS)

Custom baremetal firmware guarantees nanosecond-level timing precision by eliminating scheduling overhead.

ANCHOR & TRANSCEIVER

Anchor Processing Unit

BOARD: NUCLEO-U575ZI-Q
CORE: STM32U575

Wi-Fi Expansion Profile

BOARD:X-NUCLEO-67W61M1
INTERFACE: Shared SPI1 (via Arduino Uno V3 headers)

Shared UWB Transceiver Profile

MODULE: BU03 Module

TRANSCEIVER: Qorvo / Decawave DW3000

PHY CH / RATE: Channel 5 @ 6.8 Mbps

PREAMBLE: 128 Length

Tag Interfacing & Wiring

STM32WBA → BU03

SPI1 Interface

PA15	MOSI (Data Out)
PB3	MISO (Data In)
PB4	SCK (Serial Clock

Control & Debug

PA12	CS (Chip Select Active Low)
PA11	RST (Hardware Reset)
PB14	IRQ (EXIT 14 Notification)
UART1	TX:PB12 RX:PA8

CRITICAL CONFIGURATION: Real-time RX/TX event notifications are handled via the EXTI14 interrupt line on PB14. The debug console operates over UART1 at 115200 baud.

Anchor (Hub) Interfacing & Wiring

STM32U5⟶ BU03 & X-NUCLEO-67W61M1

SPI2 Interface : BU03 and STM32U5

PB15	MOSI (Data Out)
PB14	MISO (Data In)
PB13	SCK (Serial Clock

Control & Debug

PA4	CS (Chip Select Active Low)
PB2	RST (Hardware Reset)
PF12	IRQ (EXIT14 Notification)
PF11	WAKE

❖ We are stacking the X-NUCLEO-67W61M1 Wi-Fi expansion board directly onto the STM32 Nucleo's Arduino Uno V3 connectors to establish a multi-slave SPI1 bus layout.

Execution Trace & Validation

UART console output logs verifying the system ranging loops

System Validation Narrative

System validation is confirmed via UART serial output on the Tag node, which traces successful initialization and active ranging.

The boot sequence verifies the SPI connection integrity by reading the DW3000 Device ID (`0×DECA0302`).

Once initialized, the system enters a continuous ranging loop. The console actively logs events confirming that the Tag successfully receives broadcast polls and accurately triggers responses.

UART CONSOLE / TAG_VAL_TRACE

```
STM32 Baremetal UWB TAG Started

BU03 Device ID test: 0×DECA0302
Tag (STM32) Ready! Listening for polls...
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: 1
```

```
=====================================
         I AM ANCHOR 1
=====================================
Connecting to STM32 Wi-Fi Hub (OPEN/STATIC IP)...
.................................................
Connected to Hub Wi-Fi!
IP Address: 10.19.96.11
BU03 Device ID: 0xDECA0302
DEVICE ID: deca0302
PLL is locked..
PGF cal complete..
=====================================
  ANCHOR 1 READY (UDP CLIENT)
=====================================
ANCHOR 1 DIST: 0.28 m
ANCHOR 1 DIST: 0.30 m
ANCHOR 1 DIST: 0.29 m
ANCHOR 1 DIST: 0.28 m
ANCHOR 1 DIST: 0.29 m
ANCHOR 1 DIST: 0.30 m
ANCHOR 1 DIST: 0.31 m
ANCHOR 1 DIST: 0.33 m
ANCHOR 1 DIST: 0.34 m
ANCHOR 1 DIST: 0.34 m
ANCHOR 1 DIST: 0.34 m
```

```
=====================================
         I AM ANCHOR 2
=====================================
Connecting to STM32 Wi-Fi Hub (OPEN/STATIC IP)..
.
Connected to Hub Wi-Fi!
IP Address: 10.19.96.12
BU03 Device ID: 0xDECA0302
DEVICE ID: deca0302
PLL is locked..
PGF cal complete..
=====================================
  ANCHOR 2 READY (UDP CLIENT)
=====================================
ANCHOR 2 DIST: 0.52 m
ANCHOR 2 DIST: 0.52 m
ANCHOR 2 DIST: 0.52 m
ANCHOR 2 DIST: 0.51 m
ANCHOR 2 DIST: 0.51 m
ANCHOR 2 DIST: 0.51 m
ANCHOR 2 DIST: 0.50 m
ANCHOR 2 DIST: 0.50 m
ANCHOR 2 DIST: 0.50 m
ANCHOR 2 DIST: 0.51 m
ANCHOR 2 DIST: 0.59 m
ANCHOR 2 DIST: 0.63 m
ANCHOR 2 DIST: 0.75 m
```

```
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: E
[TAG] Responded to anchor: 1
[TAG] Responded to anchor: 2
[TAG] Responded to anchor: 1
```

```
HUB distance TAG : 0.55 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.57 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.60 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.51 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.62 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.56 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.59 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.53 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.57 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.60 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.56 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.53 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.49 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.59 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.31 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.46 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.47 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.51 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.49 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.57 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.58 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.88 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
HUB distance TAG : 0.59 m : ANCHOR 1: 0.61 m : ANCHOR 2: 0.34 m
```

Web-Dashboard